

Tutotal 5 Solution:

1. Both sets have the same elements. Order does not matter in sets. $E = F$ So, **the sets are equal sets.**

2. $n(G)=4, n(H)=4$. Since both sets have the same number of elements, they are **equivalent sets.**

3(a) Subsets of A are: $\emptyset, \{1\}$ So, **$P(A) = \{\emptyset, \{1\}\}$**

(b) Subsets of A are: $\emptyset, \{a\}, \{b\}, \{a, b\}$ So, **$P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.**

(c) Subsets are: $\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$ So, **$P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$.**

4. Number of elements in power set of a set having n elements $= 2^n$, Here, $n = 4$ So, $2^4=16$.

5. Let: T = people who like Tea, C = people who like Coffee. Total people = 60.

(a). People who like only Tea are those who like Tea but not Coffee.

$$\text{Only Tea} = n(T) - n(T \cap C) = 35 - 15 = 20.$$

(b). People who like only Coffee are those who like Coffee but not Tea.

$$\text{Only Coffee} = n(C) - n(T \cap C) = 30 - 15 = 15.$$

(c). People who like at least one of the two (Tea or Coffee)

$$n(T \cup C) = n(T) + n(C) - n(T \cap C) = 35 + 30 - 15 = 50.$$

(d). People who like neither Tea nor Coffee

$$\text{Neither} = \text{Total people} - n(T \cup C) = 60 - 50 = 10.$$

6.

(i). Let T, C, J represents the sets of students who likes Discrete, Calculus, Java respectively. Now, applying Principle of Inclusion Exclusion:

$$|T \cup C \cup J| = |T| + |C| + |J| - |T \cap C| - |C \cap J| - |T \cap J| + |T \cap C \cap J|$$

$$\text{Given } |T \cup C \cup J| = 200$$

$$200 = 120 + 90 + 70 - 40 - 30 - 20 + x$$

$$200 = 190 + x \Rightarrow x = 10$$

Now, students who like exactly two:

$$\begin{aligned} & (T \cap C - T \cap C \cap J) + (C \cap J - T \cap C \cap J) + (T \cap J - T \cap C \cap J) \\ &= (40 - 10) + (30 - 10) + (20 - 10) = 30 + 20 + 10 = \boxed{60} \end{aligned}$$

(ii).

Let A, B, C represents the sets of integers divisible by 4, 6, 10 respectively. Now, applying Principle of Inclusion Exclusion:

- A : divisible by 4 $\rightarrow 500/4 = 125$
- B : divisible by 6 $\rightarrow 500/6 = 83$
- C : divisible by 10 $\rightarrow 500/10 = 50$

Pairwise intersections:

$$|A \cap B| = 500/12 = 41$$

$$|A \cap C| = 500/20 = 25$$

$$|B \cap C| = 500/30 = 16$$

Triple intersection:

$$|A \cap B \cap C| = 500/60 = 8$$

Total divisible by at least one:

$$= 125 + 83 + 50 - 41 - 25 - 16 + 8 = 184$$

Exclude numbers divisible by all three:

$$184 - 8 = \boxed{176}$$

Solution of 7 and 8.

$$(2) A = \{1, 2, 3, 4\}, B = \{1, 3, 5\}$$

$$(a) A \cup B = \{1, 2, 3, 4, 5\}$$

$$(b) A \cap B = \{1, 3\}$$

$$(c) A - B = \{x | x \in A \text{ and } x \notin B\} = \{2, 4\}$$

$$(d) B - A = \{x | x \in B \text{ and } x \notin A\} = \{5\}$$

$$(e) A \oplus B = (A - B) \cup (B - A) = \{2, 4\} \cup \{5\} = \{2, 4, 5\}$$

$$(3) X = \{1, 2, 3, 4, 5, 6, 7\}$$

The partition of a set X is the collection

$P(X) = \{A_i | A_i \subseteq X\}$ satisfying the following properties:

i) all A_i 's are non-empty.

ii) $A_i \cap A_j = \emptyset, \forall i, j$

iii) $\bigcup_i A_i = X$

So, only S_2 is partition.

S_1 does not satisfy ii) condition.

S_3 does not satisfy iii) condition.

Solution of question 9 and 10

transitive and anti-symmetric.

⑨ $n(D) = 36$, $n(A) = 12$, $n(S) = 18$,
 $n(D \cup A \cup S) = 45$, $n(D \cap A \cap S) = 4$

No. of students who got two medals
 $= n(A \cap D) + n(D \cap S) + n(A \cap S) - 2n(A \cap D \cap S)$

We know

$$n(A \cup D \cup S) = n(A) + n(D) + n(S) - n(A \cap D) - n(A \cap S) - n(D \cap S) + n(A \cap D \cap S)$$

$$45 = 36 + 12 + 18 - n(A \cap D) - n(A \cap S) - n(D \cap S) + 4$$

$$\Rightarrow n(A \cap D) + n(A \cap S) + n(D \cap S) = 25$$

$$\text{So required answer} = 25 - 2 \times 4 = 17$$

⑩ Required answer is

$$n(A \cap D) + n(D \cap S) + n(A \cap S) - 3n(A \cap D \cap S)$$

$$= 25 - 3 \times 4 = 13$$

= No. of students who got exactly two medals.